Department für Wirtschafts- und Rechtswissenschaften

Masterarbeit

Demand Estimation with Unstructured Product Data:
Evidence from Amazon's Toothpaste Market

Burak Uzun

[Tag]. [Monat] 2026

1. Gutachter: Prof. Dr. Cristian Huse
2. Gutachter: Thomas Handke

Danksagung

[Danksagung hier einfügen.]

Erklärung

Ich erkläre, dass ich diese Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel verwendet habe.

Burak Uzun
Matrikelnummer 6494877
Oldenburg, den [TT]. [Monat] 2026

Abstract

Merger simulation hinges on correctly measuring substitution patterns, yet the nesting structures used in practice are typically built from coarse labels such as brand identity. This thesis asks whether product representations extracted from unstructured listing content – images and text encoded with CLIP – define substitution patterns better than brand labels do. Using an ASIN-by-quarter panel of the Amazon US toothpaste market around Colgate-Palmolive's 2020 acquisition of the natural-brand hello, three nested-logit demand models are estimated via the Berry (1994) linearisation that differ only in how nests are assigned: no nesting, brand nesting, and nesting by k-means clusters of CLIP image-and-text embeddings. CLIP-based nesting fits demand best and cuts the share of economically implausible negative implied marginal costs from 8.6% to 1.7%. Because CLIP places hello and Colgate in different clusters, the implied diversion between the merging parties is small: the simulated merger price effect for hello falls from +1.23% under plain logit to +0.40% under CLIP nesting. Placebo tests confirm that any cross-firm nesting can repair the cost side; what is distinctive to CLIP is that it also fits demand better than 95% of random partitions and yields a merger counterfactual free of the spurious diversion those partitions introduce. The results suggest embedding-based nests are a practical, replicable tool for antitrust analysis when conventional product characteristics are unavailable.

Zusammenfassung

Fusionssimulationen hängen entscheidend davon ab, Substitutionsmuster korrekt zu messen; die in der Praxis verwendeten Nest-Strukturen beruhen jedoch meist auf groben Merkmalen wie der Markenzugehörigkeit. Diese Arbeit untersucht, ob Produktrepräsentationen aus unstrukturierten Angebotsinhalten – mit CLIP kodierte Bilder und Texte – Substitutionsmuster besser abbilden als Markenlabels. Auf Basis eines ASIN-Quartals-Panels des US-amerikanischen Amazon-Zahnpastamarktes rund um die Übernahme der Naturkosmetikmarke hello durch Colgate-Palmolive im Jahr 2020 werden drei Nested-Logit-Nachfragemodelle mittels der Berry-(1994)-Linearisierung geschätzt, die sich ausschließlich in der Nest-Zuordnung unterscheiden: ohne Nester, Marken-Nester und Nester aus k-Means-Clustern der CLIP-Einbettungen. Die CLIP-basierte Nest-Struktur weist die beste Anpassung auf und senkt den Anteil ökonomisch unplausibler negativer implizierter Grenzkosten von 8,6 % auf 1,7 %. Da CLIP hello und Colgate unterschiedlichen Clustern zuordnet, ist die implizierte Diversion zwischen den Fusionsparteien gering: Der simulierte Preiseffekt der Fusion für hello sinkt von +1,23 % (einfaches Logit) auf +0,40 % (CLIP-Nester). Placebo-Tests bestätigen, dass jede firmenübergreifende Nest-Struktur die Kostenseite reparieren kann; CLIP zeichnet sich dadurch aus, dass es zusätzlich die Nach-

frage besser anpasst als 95 % zufälliger Partitionen und eine Fusionssimulation liefert, die frei von der durch solche Partitionen erzeugten Scheindiversion ist. Die Ergebnisse legen nahe, dass einbettungsbasierte Nester ein praktikables, reproduzierbares Instrument der Wettbewerbsanalyse sind, wenn klassische Produkteigenschaften nicht verfügbar sind.

Contents

List of Figures

List of Tables

1 Introduction

1.1 Motivation

Competition authorities are encountering mergers and analyzing their effects on the market. These mergers typically take the form of an established firm in the market acquiring a relatively small and rapidly growing brand. In an automotive market merger, regulators can observe product characteristics—such as horsepower, fuel consumption, and brand influence—as widely expected, and model substitution effects accordingly. In the case of consumer goods, however, the situation is somewhat different; differentiation here may relate to product names, product descriptions, and packaging/visual elements. It is essential to include these latent factors in the model. However, the primary tools for merger simulation have been developed for structured data labels. Nested Logit models (McFadden, 1978) typically estimate substitution effects by assigning nested categories based on brand names or manually selected segments. When these coarse observable labels misrepresent how consumers perceive products, the resulting substitution patterns and simulated price effects carry that misclassification forward, a problem that has been noted in the merger simulation literature as a source of systematic bias (Huang et al., 2008).

Another practical challenge is estimating the price effect in a robust manner. This requires variables on the external cost side (IV); however, online retail data rarely includes this variation. While input cost series vary only over time and are absorbed by quarter fixed effects, Hausman-style competitor price instruments (Hausman, 1996) show correlation across markets due to demand shocks. These instrument limitations compound the misspecification problem, and the difficulties caused by an unspecified demand model are well-documented in the literature, motivating later approaches that recover demand structure from product characteristics directly. Underestimating demand elasticity can lead to overestimated Bertrand markups and, consequently, negative marginal costs recovered from the demand side.

1.2 Research Objectives

This paper examines whether the differentiation structure of products can be extracted from unstructured data, and whether doing so produces more plausible merger price effects. We can describe this as an empirical study centered around three research questions.

First, can nests constructed from product images and text/descriptions identify substitution patterns as well as—or better than— the nests derived from brand identities? Second, to what extent does the estimated merger price effect change when the nesting rule is altered? Third, in a scenario where the missing instrument problem and endogeneity lead to an underestimation of substitutability, resulting in implausibly large markups and, consequently, negative marginal costs, do latent factors derived from unstructured data yield a more plausible economic estimate?

1.3 Approach and Contribution

The inability to identify a valid instrument was the primary problem encountered in this paper, which we overcame through a methodological change. To overcome this lack of an instrument and establish a valid model, price elasticity for CPG was calibrated using a meta-analytic finding (Bijmolt et al., 2005).

Another empirical aspect worth noting in the data is Colgate-Palmolive's acquisition of hello Products LLC, an oral care brand that positions itself as "naturally friendly" in the market. The deal, which was officially announced on January 23, 2020, was completed on January 31, 2020, for $351 million (Colgate-Palmolive, 2020). This provides a clear event date on a quarterly basis (2020Q1) and is included in the data. The ASIN-by-quarter panel data spanning from Q1 2018 to Q3 2022 is one of the empirical advantages enabling us to validate our model using this merger event and test it against simulation.

1.4 Related Literature

This study and its contributions are primarily grounded in the literature under three main headings: discrete-choice demand modeling and its application in merger analysis, where standard logit models struggle to capture realistic substitution patterns; the endogeneity problem in demand models, which biases price coefficient estimates; and the identification and calibration of the price variable, as well as the use of product representations in economic modeling and consumer choice studies, where measurement choices directly shape demand estimates.

1.4.1 Discrete Choice Demand and Merger Simulations

The demand framework is derived from the discrete choice literature. The literature shows that logit models can be linearized by inverting market shares (Berry, 1994). From there, under Bertrand competition, first-order conditions can be used to invert observed prices and recover marginal costs. The well-known BLP (Berry et al., 1995) generalized this framework to random-coefficient logit models, which allows for more flexible substitution patterns and is now standard in merger simulation. However, the resulting complex modeling (continuous consumer heterogeneity across product characteristics) made the equation multidimensional and rendered inversion impossible. To overcome this "curse of dimensionality," a contraction mapping (a fixed-point iteration algorithm) was introduced. This algorithm numerically inverted market share to recover the mean product utility (δ). Thus, once the mean utility was recovered, the effect of unobserved product quality (ξ) could be isolated as a linear residual. Although standard logit model is computationally quite advantageous over Mixed Logit, it suffers from a major drawback, the Independence of Irrelevant Alternatives (IIA) property, which motivates the use of nested logit and random-coefficient models instead. This model is based on the assumption that consumers substitute between products purely in proportion to their market shares, without considering the products' actual characteristics, which poses serious practical difficulties. While Random Coefficients models break this assumption at the aggregate level by mixing over consumer types, nested logit models relax this rigid assumption by grouping similar products (allowing the unobserved components of utility to be correlated within nests) (Cardell, 1997). Comparing RCNL (Random Coefficient Nested Logit), RC (Random Coefficient Logit), and Nested Logit (NL) models using the EU automobile market, Grigolon and Verboven (2014) found that although nested and mixed/RC logit models can produce different substitution patterns, they produce very similar price effect predictions on merger effects, nested logit even more precise to the observed effect with a better fit (NL χ^2 = 30.61 against RCNL and RC χ^2 = 423.84 against RCNL). More directly relevant here, they also concluded that segmentation via nesting captures discrete market segmentation that continuously measured product characteristics do not capture, which is what makes the nesting structure itself consequential. Based on this logic, it can be argued that choice of nesting structure—the primary focus of our study—a binding constraint. More specifically, nesting choices depend on firm-specific priors (brand, category, or observed characteristics), and the resulting substitution patterns are only as good as those priors. The basic framework for the practical application of these models is outlined in Nevo (2000a). A comprehensive application of the method—encompassing all structural stages such as demand forecasting, margin recovery, and the analysis of competitive scenarios—was successfully demonstrated by Nevo (2000b) in the context of the ready-to-eat breakfast cereal sector, a fast-moving consumer good similar to the toothpaste market.

The merger simulation employs the methodologies developed by Werden and Froeb (1994) and Nevo (2000b) to calculate pre-merger marginal costs, update the ownership structure, and recalculate Bertrand-Nash prices using the damped fixed-point iteration method examined by Morrow and Skerlos (2011), making them a natural fit for a differentiated-product market setting where Bertrand competition is the assumed pricing model. This method enables competition analysis by simulating changes in post-merger market prices. The PyBLP routines developed by Conlon and Gortmaker (2020) form the basis for validating the Bertrand-Nash equilibrium price calculations in this study, confirming that the merger simulation follows current best practices. But can such simulations actually predict the effects of a merger on the market in practice? Various studies (Peters, 2006; Björnerstedt & Verboven, 2016) have shown that in retrospective merger analyses, simulations can predict actual market movements to a large extent. For example, in the study by Björnerstedt and Verboven, the actual 39.7% price increase in the paracetamol segment in Sweden was predicted quite closely (+41.1% under the constant-expenditures nested logit). In addition to this, the placebo analysis in Section 4.6 brings the same approach to a single application—not predictions against realized prices, but the preferred nesting within the space of alternative structures.

1.4.2 Identification and Calibration of Price Sensitivity

In applied industrial organization, price endogeneity is a classic problem (Berry, 1994; Berry et al., 1995), quite common when estimating demand from aggregate market data. In its simplest definition, price endogeneity refers to the correlation between a product's price and the error term in the model—a term that is unobservable to the econometrician. More specifically, since profit-maximizing firms set higher prices for unobserved, higher-quality products, the price variable becomes correlated with unobserved demand shocks (Nevo, 2000a, 2001). Instrumental variables are used to address price endogeneity. However, in this study, the price is not instrumented but is calibrated. In line with the rationale explained in the introduction, well-known solutions such as classical cost shifters, BLP-type differentiation instruments, or other market prices introduced by Hausman (1996) rely on exclusion restrictions that are difficult to justify in a single-category panel dataset on a national platform. Therefore, the price coefficient has been fixed to an exogenous target value instead. This target elasticity value of -2.62 was taken from the study by Bijmolt et al. (2005), which conducted a meta-analysis of 1,851 own-price elasticity estimates derived from 81 studies and reported an average value of -2.62. Calibrating price sensitivity to an exogenous elasticity when appropriate cost-oriented instruments are unavailable, as is the case in a single-category national panel where cost shifters and cross-market price variation are absent, is a standard approach in applied antitrust simulations, as pioneered in the logit merger calibrations by Werden and Froeb (1994). Furthermore, this approach provides transparency; because the assumption is not hidden behind first-stage estimates but consists of a single number that is directly subject to scrutiny and the sensitivity analysis in Appendix C. Draganska and Jain (2006) also conducted structural demand forecasting for differentiated fast-moving consumer goods (CPG) lines and reported their own price elasticities within the range used to compare the values calibrated in this study.

1.4.3 Representation Learning and Unstructured Data in Demand Estimation

A growing body of literature is replacing directly observed product characteristics with vector representations (learned representations) derived from artificial intelligence and machine learning in demand estimation. In this tradition, Bajari et al. (2015) compared machine learning algorithms such as Random Forest, SVM, and LASSO with traditional econometric models using scanner panel data and demonstrated that these new methods significantly improve out-of-sample demand forecasting performance, particularly as the number of variables increases. Rather than labeling the physical characteristics of products one by one, the researchers automatically generated product features by analyzing unstructured product text descriptions using a bag-of-words model; this method constitutes one of the theoretical foundations in the literature for the use of textual data from Amazon product listings in demand models. In another consumer basket study, Ruiz et al. (2020) developed a generative probabilistic model they named "SHOPPER" to analyze large-scale market basket data. Just as word embeddings learn the meanings of words based on their positions within sentences, this model focuses on which products are purchased together rather than on direct product attributes, thereby learning "latent attributes" and vector representations for each product. These representations, derived directly from the data, not only cluster products but are also successfully used to accurately predict changes in demand under price counterfactuals and to identify substitution and complementarity relationships between products. In another study, Magnolfi et al. (2025) constructed a low-dimensional product-space embedding through crowd-sourced surveys and triplet comparisons in scenarios where product features are unobservable. When applied to the ready-to-eat breakfast cereal sector, this method demonstrated that the resulting embedding distances capture consumers' substitution patterns and cross-price elasticities in a far more consistent and rational manner than traditional models based on standard nutritional values (sugar, fiber, etc.). Finally, the study that served as the primary source of inspiration for this work—Compiani et al. (2026)—demonstrated that by extracting embeddings from product images and text and incorporating them into a mixed logit (random coefficient logit) model, it was able to capture substitution patterns in unstructured data across 40 Amazon categories that attribute-based models had missed. In this study, however, we leverage these embeddings specifically to construct the hierarchical groupings within a Berry (1994)-style nested logit model, a choice that aligns with our goal of balancing interpretability and computational efficiency for merger-review applications. This approach provides a linear, transparent, and predictably fast processing pipeline. This is precisely what merger-review processes, where speed and auditability are priorities, require. The embedding technology (CLIP) used in this study was developed by Radford et al. (2021); this architecture's contrastive image-text training makes visual and textual representations directly comparable and enables the construction of a single shared product space from both modalities.

1.4.4 Positioning

Our main contribution to the literature lies at the very heart of the main topics discussed in the three sections covered in this chapter. The study borrows the nested-logit inversion and Bertrand counterfactual analysis from the demand and antitrust literature, and a controllable external flexibility value—rather than a fragile first-stage estimate—from the calibration/identification literature. Finally, from the representation learning literature, it adopts the idea that a product's position in a learned embedding space is, in itself, an economic characteristic. The main contribution of this study lies in the way it integrates all these elements. Here, the embedding structure is incorporated into the model as a regressor, and while its significance has been measured, it is primarily designed and examined as a nesting mechanism, as mentioned.

Furthermore, this embedding has been subjected to the placebo tests described in Section 4.6; these tests provide a check for internal validity that is often overlooked in representation learning-based demand studies, which typically stop at out-of-sample fit or counterfactual price simulations without testing whether the learned structure actually reflects meaningful economic groupings. The resulting contribution is that a connection between modern representation learning and the nested logit merger simulation toolkit used by antitrust practitioners could evolve as a method that regulators can count-on when needed.

1.5 Structure

In this study, Section 2 describes the industry setting, the two data sources and the construction of the estimation panel. Section 3 sets out the methodology: the CLIP embedding pipeline that produces the nests, the product space it yields, and the Berry (1994) nested-logit estimator with its calibration and counterfactual machinery. Section 4 presents the analytical results – coefficient estimates, recovered marginal costs, the merger counterfactual, and the placebo and brand-attribution batteries. Section 5 discusses the limitations under which the findings should be interpreted, and Section 6 concludes.

2 Data

2.1 The hello–Colgate Acquisition

hello Products LLC (founded in 2012), a “natural” oral care brand targeting a natural friendly customer audience with fluoride-free options, charcoal and coconut oil based product lines, and a distinct brand voice, joined the Colgate-Palmolive family in January 2020 for a cash consideration of $351 million (Colgate-Palmolive, 2020). This merger case serves as an ideal “laboratory” for the empirical analyses in this study in three respects. First, the timing of the acquisition coincides exactly with the start of Q1 2020. Second, the acquisition of a smaller and differentiated brand (hello) by the market’s largest established leader (Colgate) ensures that price implications are directly based on the interproduct substitution relationships that are the focus of this study. The third and final reason is that both brands have a broad sales network on the Amazon platform, where product details can be directly observed.

2.2 Data Sources

The dataset forming the basis of the analyses in this study involved a very lengthy and intensive manual data manipulation process, clearly illustrating the accessibility constraints of research data in the online retail sector. The process consisted of combining two data sources. For household-level purchase history, we used the crowdsourced “Open E-commerce 1.0” dataset [Berke et al., 2024], distributed via Harvard Dataverse, which contains over 1.8 million transaction records from 5,027 U.S. Amazon consumers. This dataset has been validated by its developers using Amazon’s publicly available sales data; a Pearson correlation coefficient of $r = 0.978$ ($p < 0.001$) was found between total spending in the panel and Amazon’s published revenue [Berke et al., 2024]. The toothpaste purchase records extracted from the dataset for this study cover the period from January 2018 to March 2023 and include order timestamps, prices, and household ID numbers. Although household demographic characteristics are present in the source data, a Random Coefficient Logit model incorporating these variables effectively into the analysis was not constructed.

2.3 Sample Construction

The estimation sample was constructed in three main steps, each defined by a practical constraint:

1. **Choice-set restriction (167 ASINs).** The population of toothpaste ASINs was restricted to products purchased at least five times during the sample period. An unrestricted selection set containing thousands of rarely purchased alternatives was deemed computationally infeasible and produced singular design matrices; the five-purchase threshold filters out noise-like listings while retaining all products in the sample that contain meaningful information about revealed preferences.

2. **Brand grouping (12 groups).** Brands accounting for more than 2% of purchases each were evaluated as separate groups on their own (Colgate, Crest, SENSODYNE PRONAMEL, Sensodyne, Tom's of Maine, Arm & Hammer, hello, Parodontax, APAGARD, JASON, and Orajel); while all remaining smaller brands have been combined into a single "Other" category. The eleven groups listed represent 87% of total purchases.

3. **Panel filters (159 ASINs, 1,164 observations).** purchases are aggregated at the ASIN × quarter cell level. The forecast window ends in Q3 2022 (due to incomplete coverage of subsequent quarters), and ASINs were required to appear in at least three quarters to be included in the sample; this prevents products with market shares close to zero from dominating the sample due to sampling error. The resulting panel dataset covers the period from 2018Q1 to 2022Q3 (19 quarters, 159 ASINs, and 1,164 observations).

Market shares were derived from purchase counts based on revealed preferences. Market size was defined as $M_t = \lambda \cdot \sum_j q_{jt}$ with $\lambda = 13.93$, indicating an approximate 93% share of outside goods (Huang et al., 2008). Assuming that the consumer's primary decision-making mechanism (discrete choice) on e-commerce platforms is not "how many grams should I buy?" but rather "which product (ASIN) should I choose?" By treating package size as a control variable in the model—and thus already accounting for the utility effect of weight differences within the mean utility equation—it was decided to calculate market shares based on purchase count rather than grams. One can see robustness checks of market size in Appendix C.

Regardless of these sample filters, the shared embedding space in Section 3.1 was trained on 155 of the 167 ASINs that had both a valid image and a valid text embedding; the 12 ASINs with only a text embedding were projected onto this space and retained their positions in the prediction sample. Summary statistics for the prediction panel are presented in Table 1.

Table 1: Panel Summary Statistics (Estimation Sample)

Variable	Mean	Median	SD	Min	Max
Price (USD)	9.20	8.47	5.34	1.35	41.75
Quarterly units sold	2.17	1.00	1.96	1.00	18.00
Market share (x1,000)	1.093	0.753	1.036	0.335	9.802
Package size (g)	320	259	223	43	1179
Purchasing households per cell	2.06	1.00	1.85	1.00	17.00

Notes:

1,164 ASIN x quarter observations; 159 ASINs; 19 quarters (2018Q1–2022Q3); 12 brand groups. Price is the quarterly mean deflated transaction price per ASIN. Market share uses market size M_t = (1 + 13.93) x total quarterly purchases (outside-good share approx. 93%).

3 Methodology

3.1 From Listings to Nests: Embeddings, Joint PCA, and Clustering

Each product j is represented by concatenating the list image with the title and benefit description in sequence. CLIP (Contrastive Language-Image Pre-training; Radford et al., 2021) provides a visual encoder (f_I) and a text encoder (f_T) that are trained contrastively to ensure that matched image-text pairs are positioned close to each other in a shared representation space. In the specific model variant used here (clip-vit-base-patch32), both encoders map their inputs into $\mathbb{R}^{512}$, and the outputs are L2-normalised:

$$\mathbf{e}_j^I = \frac{f_I(\text{image}_j)}{\|f_I(\text{image}_j)\|}, \qquad \mathbf{e}_j^T = \frac{f_T(\text{text}_j)}{\|f_T(\text{text}_j)\|}.$$

In this study, the functionality of the CLIP model lies in its training objective. The CLIP model consists of two distinct network architectures: one is the more well-known text transformer (Vaswani et al., 2017), and the other is the ViT-B/32 Vision Transformer (as used in this study). This vision transformer processes each image sequentially by dividing it into 32x32-pixel patches, similar to how a language model processes tokens (Dosovitskiy et al., 2021, sec. 1; the checkpoint is specified in Radford et al., 2021, sec. 2.4-2.5). Both networks are mapped to a 512-dimensional space, and model training was conducted in a contrastive manner on 400 million image-text pairs collected from the web: In each training batch consisting of N pairs, the model is trained to maximize the cosine similarity of the N correct image-caption matches while minimizing the similarity of all incorrectly matched pairs (Radford et al. 2021, sec. 2; Figure 1 summarizes this scheme). Figure 1 illustrates the objective and role of this empirical workflow.

Two characteristics of the conrtastive training objective matter directly for how product caption data is used here. First, since the product captions describe products using everyday marketing language like “activated charcoal whitening toothpaste,” the trained space encodes the vocabulary of product features that distinguish consumer goods without requiring any category specific additional training. Also, since images and texts are embedded in a single integrated space, these two modalities can be combined into a single product representation. We explain how the joint Principal Component Analysis (PCA) utilized below takes advantage of this method.

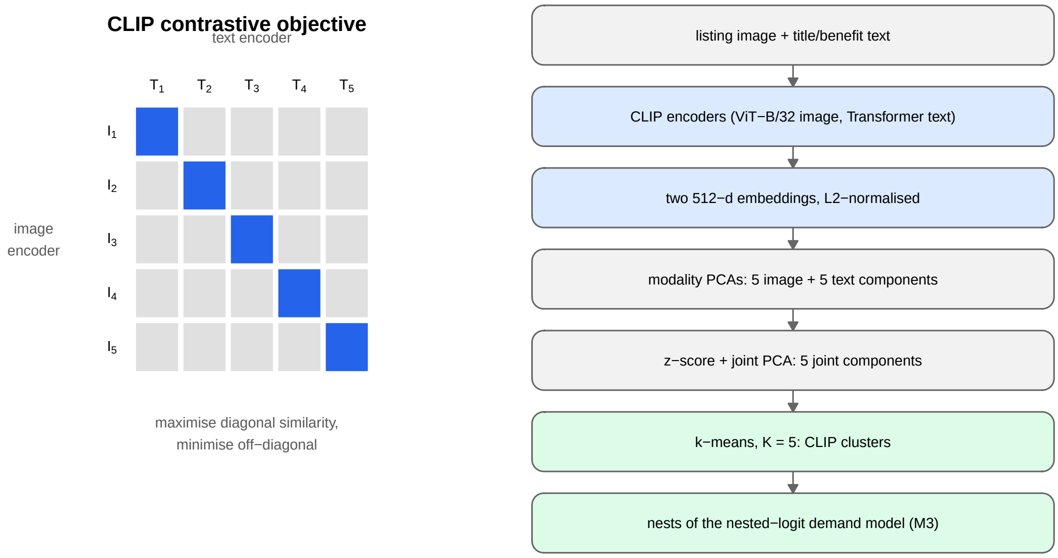

Figure 1: How CLIP embeddings are produced and used. Left: the contrastive training objective (after Radford et al., 2021). Right: the empirical pipeline from raw listing content to the nests of model M3.

Each modality's embeddings are first reduced to five modality-specific principal components (Hotelling, 1933), with each PCA fitted only on the products holding a valid embedding for that modality. This yields a ten-dimensional characteristics vector per product, $\mathbf{x}_j = (x^I_{j1},\ldots,x^I_{j5},x^T_{j1},\ldots,x^T_{j5})$.

Since these ten columns come from two separate PCA fits, and in a situation where within each modality the first component mechanically carries the largest variance, one needs to make their scales comparable. Because the image and text blocks have difference in their overall variance levels, combining them directly would let the block with the higher variance dominate the joint representation. Therefore, joint PCA is calculated on the correlation scale. Each column k is z-scored using its mean μ_k and standard deviation σ_k over the $N = 155$ products with both modalities,

$$z_{jk} = \frac{x_{jk} - \mu_k}{\sigma_k},$$

and the rotation matrix $\mathbf{W}$ is obtained from the PCA of the standardised matrix $\mathbf{Z}$. The first 5 joint principal components, $\mathbf{pc}_j = \mathbf{z}'_j \mathbf{W}_{[\cdot,1:5]}$, define the differentiation space used in the analysis. The applied standardization prevents the first two principal components (PC1, the one with the highest variance) from dominating the joint structure and ensures the creation of an overall cross-modal structure.Additionally, the fitted parameters $(\mu_k, \sigma_k, \mathbf{W})$ are saved with the replication data, which makes the projection of any product into the joint space exactly reproducible. Another indicative method allows us to numerically verify that the joint space indeed sufficiently combines the two different modalities.

To do this, we examine the fraction of its squared rotation loadings attributable to image inputs. Which is

$$\text{share}_m^I = \frac{\sum_{k \in I} w_{km}^2}{\sum_k w_{km}^2}, \qquad m = 1, \ldots, 5.$$

As Figure 2 shows, the weights of all retained joint principal components range from 48.6% to 51.4%. This confirms that the joint space is indeed multimodal.

The clustering structure was established using k-means (Lloyd, 1982), and each product is assigned to one of $K = 5$ clusters.

$$g_j = \arg \min_{g \in \{1, \ldots, K\}} \|\mathbf{pc}_j - \mathbf{c}_g\|^2,$$

with centroids $\mathbf{c}_g$ from 50 random draws, and cluster structute is the nest assignment in M3. These 5 nests explain 71.2% of joint principal components variance. One can see Appendix A, to check cluster quality diagnostics – within-cluster sum of squares and average silhouette width (Rousseeuw, 1987) – together with all downstream results for $K = 2, \ldots, 10$.

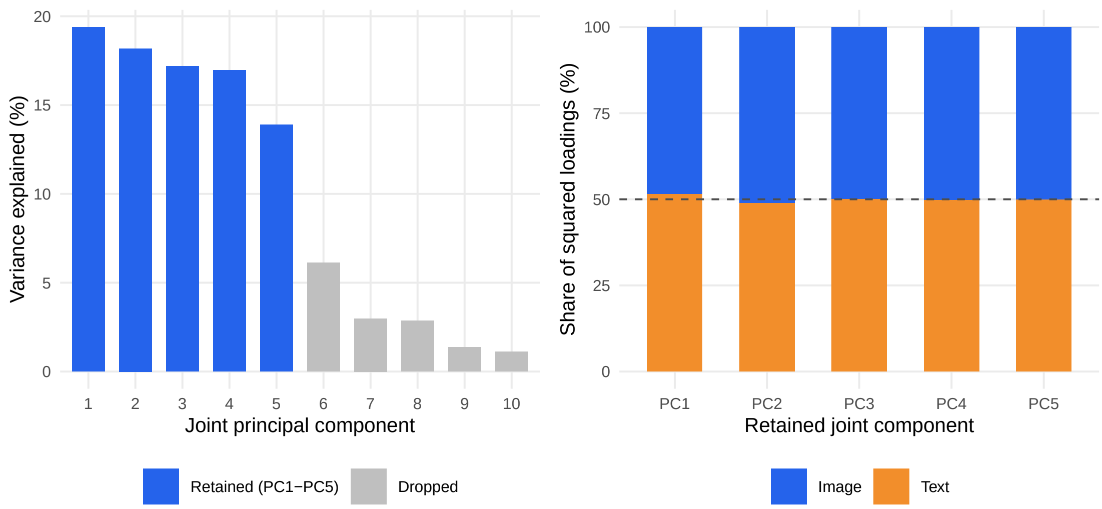

Figure 2: Anatomy of the joint embedding space. Left: variance explained by each joint principal component; the first five (blue) are retained. Right: image versus text composition of each retained component, all within 1.4 percentage points of an even split.

Product PCs are static within the observation window; since Amazon product images and descriptions change only marginally over the sample time interval, re-extracting the embeddings on a quarterly basis yields nearly identical vectors. The identifying variation is **cross-sectional**, meaning that ASINs within the same brand genuinely differ in packaging and benefit claims. This is precisely the variation captured by moving from brand-level to ASIN-level data. In addition, time-varying demand conditions are accounted for by quarterly fixed effects.

Twelve product ASINs have no image embeddings (image retrieval failed at the time of extraction) but have usable text embeddings. Instead of dropping these or imputing them with brand means, they are projected into the joint PC space from their standardized text components, with image dimensions set to the cross-sectional mean—zero in standardized space—so that $\mathbf{pc}_j = (\mathbf{0}, \mathbf{z}_j^T)' \ \mathbf{W}_{[\cdot,1:5]}$ using the same saved $(\mu_k, \sigma_k, \mathbf{W})$. These ASINs span Crest (4), Sensodyne (2), Tom's of Maine (2), hello (2), and Other (2), and all pass the three quarter inclusion filter. Because dropping these ASIN's would end up in distortion in brand level market shares and eliminate hello observations within the merger simulation interval, we decided to reject brand mean imputation approach since it would reintroduce precisely the aggregation that the ASIN level design is intended to avoid in first principle.

3.2 The Resulting Product Space

Table 2 presents descriptive statistics for CLIP clusters. The clusters identify economically meaningful differentiation segments without using price or sales data. The clusters are segmented as follows: a premium sensitivity segment (C1, dominated by GSK brands), the two mainstream incumbents (C2 Colgate, C3 Crest), Tom's of Maine (C4), and a natural/specialty segment (C5), which primarily represents "Other" brands and also includes hello. Taking into account that prices and package sizes differ significantly across clusters, we can say that the clusters capture both vertical and horizontal differentiation.

Table 2: Summary Statistics by CLIP Cluster

			Price (USD)			Mkt. Share (x1000)			Pkg. Size (g)		
Cluster	Dom. Brand	N ASINs	$\bar{p}$	$\tilde{p}$	σ_p	$\bar{s}\times 10^3$	$\tilde{s}\times 10^3$	$\sigma_s\times 10^3$	$\bar{w}$	$\tilde{w}$	σ_w
C1	Sensodyne	29	10.26	9.63	5.58	1.068	0.770	0.848	249	249	145
C2	Colgate	33	8.00	8.39	3.65	1.303	0.817	1.259	537	399	296
C3	Crest	29	8.72	6.88	4.18	1.267	0.757	1.282	375	431	128
C4	Tom's of Maine	13	9.85	9.76	2.96	1.170	0.770	0.972	374	370	94
C5	Other (hello)	55	9.46	7.86	6.84	0.839	0.670	0.736	171	141	105

Notes:
Panel: 1,164 ASIN x quarter observations, 19 quarters (2018Q1–2022Q3). Mean/Median/SD of price in USD, market share scaled x1000, package weight in grams. Dominant brand = modal brand by ASIN count within cluster.

Apart from these (price and market share data), the clusters can be directly interpreted from the Amazon listing content itself. Table 3 shows the listing text words that are most overrepresented relative to the entire category (by lift: the share of the cluster's ASINs whose title or benefit text contains the term, divided by the category-wide share). This table confirms that the clusters track not only the generic embedding geometry but also meaningful product positioning.

Table 3: Distinctive Listing Vocabulary by CLIP Cluster

Cluster	Dominant brand	N ASINs	Most over-represented listing terms (by lift)
C1	Sensodyne	29	sensodyne, parodontax, extra, bleeding, pronamel, reharden
C2	Colgate	33	optic, hydrogen, max, benefits, extreme, frosty
C3	Crest	29	scope, pro, radiant, brilliance, eraser, glamorous
C4	Tom's of Maine	13	maine, tom's, children's, silly, animals, not
C5	Other	55	sls, activated, mouth, orajel, remineralizing, anti

Notes:
Lift = share of the cluster's ASINs containing the term divided by the category-wide share. Terms appear in at least two ASINs with lift above 1.5.

Figure 3 visualizes the CLIP product space: "hello" 's listings are located in the "natural/specialty" cluster on the left side of the embedding plane, while Colgate dominates the opposite cluster region, making the low implied diversion between the merging parties easily and directly visible.

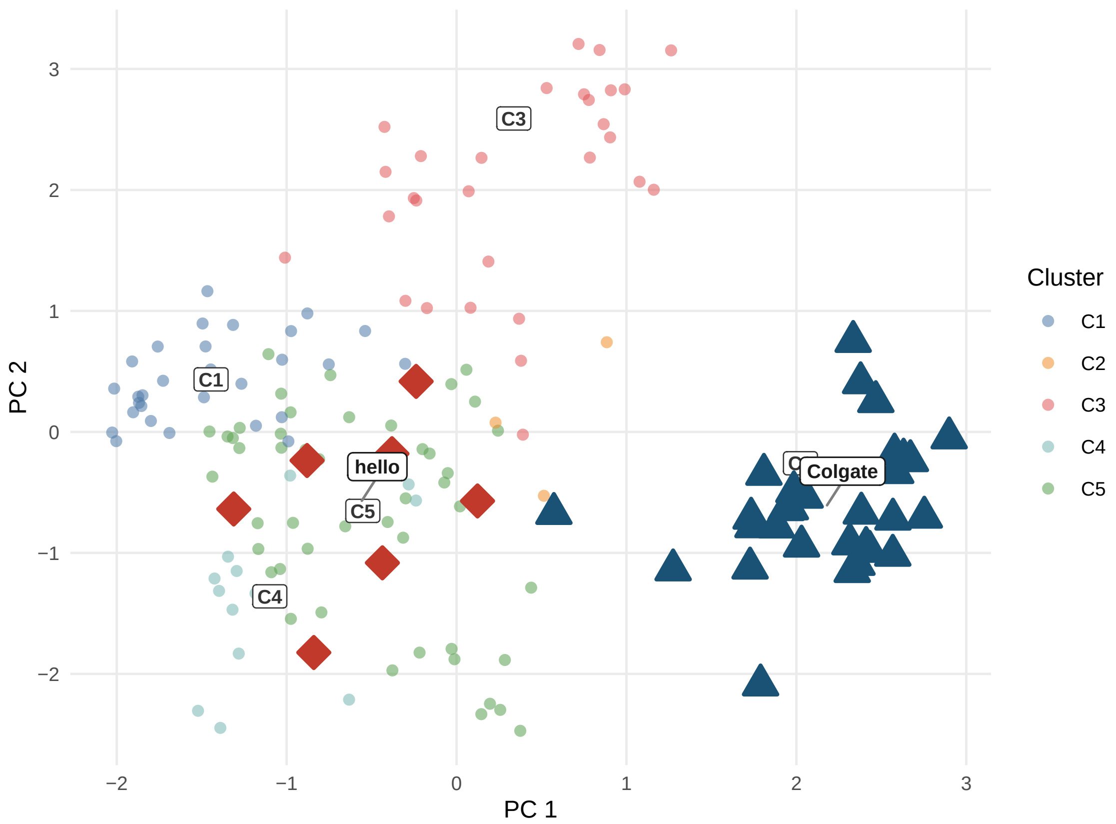

Figure 3: Product space from CLIP embeddings, each point one ASIN in the first two joint principal components. hello (red diamonds, C5) and Colgate (dark blue triangles, C2) fall in different clusters.

3.3 Berry (1994) Linearisation

All three models are estimated using the Berry (1994) nested-logit linearisation. which can be formulated as:

$$\log \frac{s_{jt}}{s_{0t}} - \rho \log s_{j|g,t} = \alpha p_{jt} + \beta' \mathbf{x}_{jt} + \mathrm{FE}_t + \xi_{jt}$$

where the share within a group is $s_{j|g,t} = s_{jt} / \sum_{k \in g} s_{kt}$, $\rho \in [0,1)$ is the nesting intensity, and FE_t represents the quarterly fixed effects absorbed through demeaning. The choice model used is based on McFadden's (1974, 1978) generalized extreme-value, and the nested logit is a closed-form member of this family. The theoretical justification for ρ is based on the variance components framework developed by Cardell (1997), and Train (2009, chs. 3–4) provides a comprehensive resource on this topic.

A detailed explanation of the aforementioned estimating equation is essential, as it can better highlight the contribution of this study, and is worth walking through.

The utility that consumer i derives from product j in nest g is defined as $u_{ijt} = \delta_{jt} + \zeta_{igt} + (1-\rho)\,\varepsilon_{ijt}$. Here, the mean utility $\delta_{jt} = \alpha p_{jt} + \beta' \mathbf{x}_{jt} + \mathrm{FE}_t + \xi_{jt}$, an i.i.d. extreme value

ε_{ijt}, and the nest-specific term ζ_{igt} from Cardell (1997), which links products within a nest together with strength ρ. We can explain the decomposition in the equation as follows: Let $D_{gt} = \sum_{k\in g} \exp\big(\delta_{kt}/(1-\rho)\big)$ be the inclusive value (log sum) of a nest; this is the expected utility a consumer would receive when choosing any product within that nest. Thereafter, GEV selection probabilities are divided into two categories: within-nest piece selection and between-nest piece selection:

$$s_{jt} = s_{j|g,t} \cdot s_{g,t}, \qquad s_{j|g,t} = \frac{\exp\big(\delta_{jt}/(1-\rho)\big)}{D_{gt}}, \qquad s_{g,t} = \frac{D_{gt}^{1-\rho}}{\sum_{g'} D_{g't}^{1-\rho}}.$$

Here, the within-nest component represents the probability that a consumer will choose a specific product from the set they have already selected, while the between-nest component represents the probability that a consumer will choose a specific nest. The final market share is represented by the product

$$s_{jt} = s_{j|g,t} \cdot s_{g,t}.$$

Now, let's assume the singleton (outside good) nest has zero-mean utility $\delta_{0t} = 0$. Then, according to equation $s_{g,t}/s_{0t} = D_{gt}^{1-\rho}$, and $\log s_{jt} - \log s_{0t} = \log s_{j|g,t} + (1-\rho)\log D_{gt}$; when we rearrange the within-nest share term ($\log D_{gt} = \delta_{jt}/(1-\rho) - \log s_{j|g,t}$). After some substitution and cancellation, the Berry (1994) linearization follows:

$$\log \frac{s_{jt}}{s_{0t}} = \delta_{jt} + \rho \, \log s_{j|g,t}.$$

Consequently, ρ enters the equation through only one observable term, $\log s_{j|g,t}$, and the nest g is observable only in terms of how that within-nest share is composed. In this case, since no term dictates how the nest structure must be formed, this ensures that any flexible nesting strategy yields a valid $s_{j|g,t}$. And it is precisely this flexibility that the empirical design of this study takes advantage of. Section 3.1 constructs g from CLIP embedding clusters instead of direct brand identity, and leaves every other term unchanged.

3.4 Price Coefficient Calibration

The problem of the lack of valid cost-side instruments, which we mentioned earlier, has forced us to find a different solution. Therefore, following the recommendation in the study by Werden and Froeb (1994) and also as a method occasionally used in the applied IO literature (DeSouza, 2009), α is calibrated to match a target median own-price elasticity. Here, the target price elasticity $\varepsilon_{\text{target}} = -2.62$, is based on the meta-analysis by Bijmolt et al. (2005, p. 141) and is consistent with analogous packaged goods.

$$\alpha = \frac{\varepsilon_{\text{target}}}{\text{median}(p_{jt}(1 - s_{jt}))} = \frac{-2.62}{8.4502} = -0.310$$

Since the calibration of α is performed under the $\rho = 0$ setting, the target -2.62 reflects the median of the logit. After introducing the nesting model, the equation inherently adjusts the price sensitivity by a factor of $1/(1-\rho^*)$. As a result, the fully calibrated α under Brand Nesting (Model 2) -3.83, while under CLIP Nesting, it has been recalibrated to -7.31. These values were calculated observation by observation, which closely align with the $1/(1-\rho^*)$ approximations of -4.0 and -7.9.

Therefore, the high price elasticities observed here reflect the granular nature of our data rather than a conflict between calibration targets. Furthermore, while the average price elasticity of -2.62—calculated based on 1,851 elasticities in the meta-analysis by Bijmolt et al. (2005)—is primarily derived from brand-level aggregates (1,218 brands, 633 SKU levels), while our data consists of observations at the SKU level. Given that individual listings face far more substitution options compared to an entire brand's portfolio, higher elasticity at the product level is expected. This approach is consistent with Draganska and Jain (2006), who report product line elasticities ranging from -2.45 to -6.25 when estimating a nested logit model using product attributes. Here, the implied median elasticities for our CLIP nesting model are slightly above this range; however, differences in the data and the strong substitution effect should be taken into account. However, in the robustness checks we conducted to assess this variation (Appendix C), we observed that the supply-side figures force the ρ^* calibration to artificially set α to -2.62 for each nest, creates compression on α and inflates Bertrand markups to unrealistic levels—even well beyond the plain-logit baseline. Consequently, it can be argued that price elasticity at the SKU/ASIN level must be higher than aggregate benchmarks for the cost side and observed product prices to stay economically plausible.

3.5 ρ^* Selection — Concentrated OLS Profile

The searching of the optimal ρ^* is based on the "concentrating out" technique. This technique was introduced to the literature by Berry et al. (1995)—famously known as BLP—and has been used for structural demand forecasting, as also famously known in the literature by Nevo (2000a). Although this approach was originally used in Random Coefficient models to isolate linear parameters from the nonlinear search space, it has also been directly integrated into the nested logit framework by Grigolon and Verboven (2014) by isolating the nesting parameter ρ^* as a nonlinear parameter.

Since the price coefficient α is already known (because it's calibrated as explained in detail in Section 3.4), the nested logit inversion yields an equation that is strictly linear in β for any given ρ, the linear product terms β can be analytically concentrated out of the linear search routine. As a result, we do not estimate/search for all parameters at once. But we reduce the problem has to a simply designed one dimension grid search for ρ.

Now the search is on a ρ on a defined grid $\{0, 0.01, \ldots, 0.90\}$, the adjusted dependent variable $\tilde{y}_{jt}(\rho)$ is formed, and the corresponding $\beta(\rho)$ vector is estimated using Ordinary Least Squares:

$$\tilde{y}_{jt}(\rho) = \log \frac{s_{jt}}{s_{0t}} - \alpha p_{jt} - \rho \log s_{j|g,t}$$

The optimized ρ^* is chosen as the final nesting parameter that minimizes the residual sum of squares of the model:

$$\arg\min_{\rho} \left\| \tilde{y}(\rho) - \mathbf{X}\hat{\beta}(\rho) \right\|^2$$

Main advantage of this approach is that it does intentionally bypass local optimum complications and solves the algorithmic convergence failure problems that frequently encountered in demand estimation scenarios (Conlon & Gortmaker, 2020).

3.6 Marginal Costs and Merger Simulation

Marginal costs are calculated from inverting pre-merger Bertrand FOCs (Berry, 1994) using the firm matrix formulation introduced by Nevo (2001):

$$\mathbf{mc} = \mathbf{p} + (\Omega_{\text{pre}} \circ \Delta)^{-1} \mathbf{s}$$

where $\Omega[j,k] = \mathbf{1}[\text{firm}_j = \text{firm}_k]$ and Δ is the demand Jacobian with elements:

$$\Delta_{jj} = \alpha s_j \left[\tfrac{1}{1-\rho} - \tfrac{\rho}{1-\rho} s_{j|g} - s_j \right], \quad \Delta_{jk}^{\text{same}} = \alpha s_j \left[-\tfrac{\rho}{1-\rho} s_{k|g} - s_k \right], \quad \Delta_{jk}^{\text{diff}} = -\alpha s_j s_k$$

The calculation of marginal cost recovery is closely related to the nest structure. If an entire nest consists of only a single firm (as in the brand nest—M2 model), Bertrand FOCs will yield the same result as the plain logit (M1). The reason for this is that markups are uniform within each firm at $1/(|\alpha|(1-S_f))$, where S_f is the firm's market share, for any value of ρ (proof in Appendix D). As a result, the plain logit and brand-nested logit models calculate exactly the same costs. For this reason, the brand-nested logit cannot generate different Lerner indices and can't correct the negative cost-side calculations. The CLIP nests that provide the solution here are able to do so using latent factors (or residuals that the brand model cannot observe) that can cut across ownership boundaries to a certain extent, beyond the variance explained by the brands.

The recovered marginal costs are calculated as the pre-merger average for each ASIN and are held constant in both counterfactuals. There are two main objectives for fixing costs to the pre-merger period: First, this choice is fully consistent with the "no-efficiencies" standard in antitrust studies (Farrell & Shapiro, 1990). Second, if the cost equation had been run backward using post-merger costs, the result we would obtain would simply be the prices we had already input into the model.

From there, we calculate two Nash equilibria for each post-merger quarter using the damped fixed-iteration (Morrow & Skerlos, 2011):

$$\mathbf{p}_{n+1} = \kappa \left[\mathbf{mc} - (\Omega \circ \Delta(\mathbf{p}_n))^{-1}\mathbf{s}(\mathbf{p}_n)\right] + (1-\kappa)\,\mathbf{p}_n$$

In the algorithm, convergence tolerance was set to 10^{-8}, while the dampening factor was set at $\kappa = 0.4$. 2,000 iterations were performed. The ownership matrix Ω_{pre} was used in the counterfactual scenario that assumed the merger did not take place. The simulation ownership matrix for the merger scenario was set to Ω_{post} (hello products given to Colgate's firm identification ID from 2020Q1). As a result, these two counterfactual calculations produced two distinct merger price impacts ($\Delta p = p^{\text{merger}} - p^{\text{no-merger}}$).

About the algorithm validation: The code implementation in R reflects the counterfactual equilibrium routines as in PyBLP (Conlon & Gortmaker, 2020). It is validated against PyBLP's compute_prices() method. In both M1 (plain logit) and CLIP nested model($\rho^* = 0.67$), the maximum absolute price has differed below 2×10^{-8} across 847 post-merger observations. Confirming the code validity.

4 Results

4.1 Demand Estimates

The demand parameters for all three models are shown in Table 4; in this table, the calibrated α value is distinguished from the estimated β and the ρ^* value obtained through profiling. Compared to the plain logit, the brand-based nesting method yields a $\rho^* = 0.35$ value with a 3.1% improvement in RSS, while the CLIP nesting method yields a $\rho^* = 0.67$ value with an 8.1% improvement in RSS. As indicated by the higher value of the profiled embedding parameter, the embedding-to-embedding substitution process is significantly more intensive when embeddings are defined on a brand-by-brand basis compared to when they are constructed in the embedding space.

Table 4: Demand Parameter Estimates — Three Models

Variable	M1: Logit (ρ=0)	M2: Brand-Nested (ρ^*=0.35)	M3: CLIP-Nested (ρ^*=0.67)
α (price)	**-0.310 (a)**	**-0.310 (a)**	**-0.310 (a)**
Package size (g)	0.0037*** (0.0003)	0.0037*** (0.0003)	0.0034*** (0.0003)
CLIP PC_1	-0.4034*** (0.0424)	-0.2869*** (0.0418)	-0.3280*** (0.0407)
CLIP PC_2	0.2033*** (0.0374)	0.2681*** (0.0368)	0.2103*** (0.0359)
CLIP PC_3	-0.1667*** (0.0386)	-0.1479*** (0.0380)	-0.2854*** (0.0370)
CLIP PC_4	0.2115*** (0.0399)	0.1574*** (0.0393)	0.2707*** (0.0382)
CLIP PC_5	-0.1652*** (0.0390)	-0.1243*** (0.0384)	-0.1745*** (0.0374)
Other brand	0.0849 (0.1327)	0.4590*** (0.1307)	0.2641** (0.1272)
Import x freight shock	0.6767*** (0.2342)	0.6709*** (0.2306)	0.7461*** (0.2245)
ρ^*	0.000 (b)	0.350 (b)	0.670 (b)
Quarter FEs	Yes	Yes	Yes
Median implied ε_{jj}	-2.62	-3.83	-7.31
RSS drop vs ρ=0	—	3.1%	8.1%

Notes:
*** p<0.01, ** p<0.05, * p<0.10. FEs by quarter. N=1,164. (a) calibrated alpha; (b) grid-searched rho*. Sections 4.2-4.3.

4.2 Own-Price Elasticities by Brand

The median own-price elasticities by brand group for each model are shown in 5. There are two notable motifs. Initially, elasticities scale with price levels in each model. Premium brands like APAGARD and SENSODYNE PRONAMEL are the most elastic as the nested-logit own-price elasticity is proportional to αp_j. Second, from plain logit to CLIP nested logit all elasticities differ by $1/(1-\rho^*)$, such increase can be explained with cluster composition: brands that share a dense embedding cluster are subject to the largest within-nest competitors. Same trend is also observable for the merging brands. Because its closest substitutes are the natural/specialty goods in its own nset rather than Colgate, hello's median elasticity increases from -1.42 under plain logit to -4.05 under CLIP nesting.

Table 5: Median Own-Price Elasticities by Brand Group

Brand	N obs.	M1: Logit	M2: Brand-Nested	M3: CLIP-Nested
APAGARD	44	-4.92	-6.62	-14.33
Sensodyne	99	-3.00	-4.27	-8.68
SENSODYNE PRONAMEL	90	-3.04	-4.42	-8.67
Colgate	**224**	**-2.84**	**-4.27**	**-8.24**
Tom's of Maine	85	-3.03	-4.38	-8.07
Other	220	-2.70	-4.09	-7.98
Crest	219	-2.11	-3.18	-6.17
Parodontax	51	-1.79	-2.54	-5.29
Arm & Hammer	21	-1.53	-1.62	-4.22
hello	**50**	**-1.42**	**-1.88**	**-4.05**
JASON	29	-1.30	-1.48	-3.77
Orajel	32	-0.86	-1.16	-2.53

Notes:
Median own-price elasticity per brand group over the full panel, using the nested-logit formula of Section 3.6. Bold = merger parties.

4.3 Diversion Ratios

The key question in the merger is what will happen to demand for hello products when there is a prices increase. To answer this, we look at the cross-price elasticities to show how price changes affect demand for different products. Table 6 presents these ratios—calculated using the formula $DR_{j\to k} = -(\partial s_k/\partial p_j)/(\partial s_j/\partial p_j)$—for "hello" products, which are categorized into fixed categories and weighted based on market share and pre-merger quarters. These diversion ratios are of vital importance in assessing the potential impact of a merger; they serve as a fundamental input in the modern effect monitoring that regulatory agencies use to evaluate the effects of mergers on competition (Conlon & Mortimer, 2021). Indeed, these ratios are directly referenced in the U.S. Horizontal Merger Guidelines, particularly in the analysis of price pressure from differentiated products (Farrell & Shapiro, 2010), underscoring just how important they are in merger decision making. (U.S. Department of Justice & Federal Trade Commission, 2010, sec. 6.1)

Table 6: Diversion from hello by Destination (percent of lost demand)

Model	To Colgate	To rest of hello's CLIP cluster	To other inside goods	To outside good
M1: Logit	1.16	1.49	3.88	93.47
M2: Brand-Nested	0.93	1.20	22.41	75.45
M3: CLIP-Nested	**0.41**	**58.14**	**8.39**	**33.06**

Notes:
Share-weighted average across hello ASINs over pre-merger quarters, from each model's demand Jacobian. Rows may not sum to 100 due to rounding.

The three demand systems tell three different stories about the same market. Under plain logit, diversion is proportional to market shares, so 93% of hello's marginal demand flows to the outside good and only 1.2% reaches Colgate. Brand nesting redirects substitution toward hello's own sister ASINs but leaves cross-brand diversion share-proportional. Under CLIP nesting, 58% of hello's lost demand diverts to the other natural/specialty products in its cluster, while diversion to Colgate falls to 0.4% – the low-diversion configuration that drives the small simulated merger effect in the next subsection.

4.4 Merger Counterfactuals

Table 7 reports the merger counterfactuals: observed prices, recovered marginal costs, the no-merger Nash equilibrium, the merger Nash equilibrium, and the implied price effect Δp for the merging parties and the largest rivals. Under plain logit, where diversion is proportional to market shares, the simulated effect for hello is +1.23%; brand nesting barely changes it (+1.28%) because hello's ASINs sit in their own brand nest and diversion to Colgate is still share-proportional across nests. Under CLIP nesting, hello (C5) and Colgate (C2) are distant in product space, diversion between the parties is small, and the simulated hello effect falls to +0.40%, with Colgate's own effect economically negligible.

Table 7: Merger Counterfactuals: hello-to-Colgate Acquisition (2020Q1)

Brand	Obs. p	MC	Nash (pre)	Nash (post)	Δp	Δp %
M1: Logit (ρ=0)						
hello	**6.43**	**3.20**	**7.10**	**7.16**	**0.064**	**1.233%**
Colgate	**8.45**	**5.16**	**8.29**	**8.30**	**0.007**	**0.104%**
Tom's of Maine	10.47	7.17	10.70	10.71	0.006	0.069%
Crest	8.87	5.60	9.18	9.18	0.000	0.000%
Sensodyne	10.45	7.17	10.05	10.05	0.000	0.000%
M2: Brand-Nested (ρ^*=0.35)						
hello	**6.43**	**3.20**	**7.10**	**7.17**	**0.065**	**1.275%**
Colgate	**8.45**	**5.16**	**8.30**	**8.30**	**0.006**	**0.104%**
Tom's of Maine	10.47	7.17	10.71	10.71	0.006	0.069%
Crest	8.87	5.60	9.18	9.18	0.000	0.000%
Sensodyne	10.45	7.17	10.07	10.07	0.000	0.000%
M3: CLIP-Nested (ρ^*=0.67)						
hello	**6.43**	**5.20**	**7.05**	**7.07**	**0.020**	**0.404%**
Colgate	**8.45**	**5.88**	**8.59**	**8.59**	**0.002**	**0.018%**
Tom's of Maine	10.47	7.19	10.72	10.72	0.002	0.021%
Crest	8.87	5.63	9.18	9.18	0.000	0.000%
Sensodyne	10.45	7.28	10.29	10.29	0.000	0.000%

Notes:

All prices USD; post-merger averages (2020Q1–2022Q3). Bold = merger parties. Nash (pre): Bertrand-Nash under pre-merger ownership. Nash (post): equilibrium under merged ownership. Dp = Nash (post) minus Nash (pre). MC inverted from pre-merger FOC and held fixed.

Figure 4 presents post-merger price trajectories for each quarter. It can be seen that the prices after the merger are differentiating "if there was no merger" scenario, especially for hello. But for Colgate, the prices look almost identical for both.

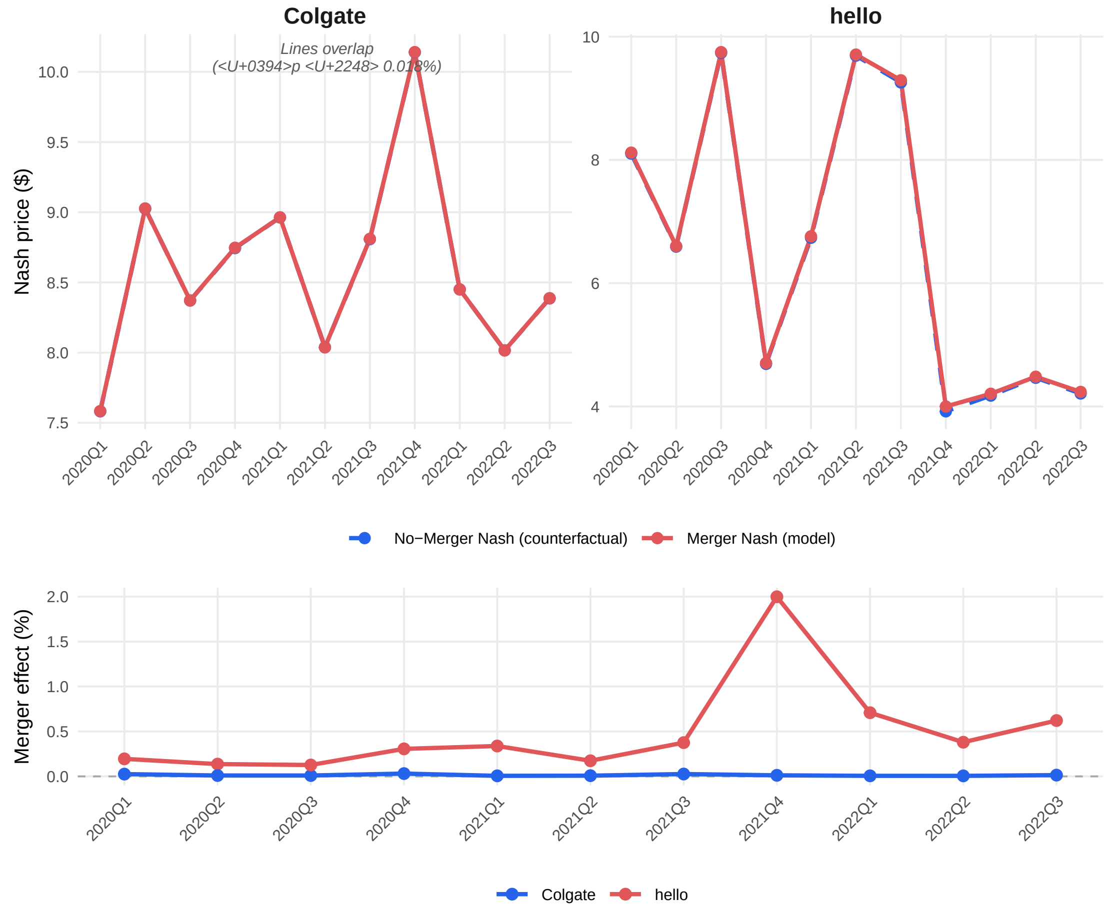

Figure 4: Merger simulation price trajectories (Model 3, CLIP-Nested, $\rho^* = 0.67$). Top: Nash prices under merged and pre-merger ownership. Bottom: the merger price effect in percent, positive for hello (red) and near zero for Colgate (blue).

4.5 Cross-Model Comparison

Comparisons of the economic plausibility of the results across the three models are summarized below. As has been mentioned repeatedly, the main reason is the inflated markups resulting from an underestimation of substitutability. The last column shows how prices affect consumers, using a special “nested logit inclusive value” formula, which was developed by Small and Rosen in 1981. It measures how much consumers wealth is reduced by certain pricing practices.Consequently, we see that in CLIP nested logit the harm to consumers is about one-third of what it would be in plain logit.

Table 8: Cross-Model Comparison: Demand Fit and Economic Plausibility

Model	ρ^*	Med. ε_{jj}	RSS drop	Neg. MC %	Med. Lerner	hello Δp	Colgate Δp	ΔCS
M1: Logit	0.00	-2.62	—	8.6%	38.5%	+1.23%	+0.104%	-25.4
M2: Brand-Nested	0.35	-3.83	3.1%	8.6%	38.5%	+1.27%	+0.104%	-26.6
M3: CLIP-Nested	**0.67**	**-7.31**	**8.1%**	**1.7%**	**28.9%**	**+0.40%**	**+0.018%**	**-8.5**

Notes:
All statistics over the full estimation panel (1,164 ASIN x quarter observations) except merger effects (post-merger quarters 2020Q1–2022Q3). Med. elasticity = median own-price elasticity. RSS drop = reduction in residual sum of squares vs rho=0 from the concentrated OLS profile. Neg. MC = share of observations with negative implied marginal cost from the Bertrand inversion (lower = more economically plausible). Med. Lerner = median (p - mc)/p. hello / Colgate Dp = average post-merger Nash price effect. dCS = mean change in consumer surplus, merger vs no-merger Nash, in cents per 1,000 consumers per quarter (nested-logit log-sum formula). Bold/green row = preferred specification.

4.6 Placebo Nesting Tests

The advantage of observed CLIP embedding nesting creates a reasonable question: is this improvement coming from *semantic content* of the CLIP embeddings, or would any random clustering that can cut across ownership boundaries at some extent, can do just as well? To test this, we have created two kinds of placebo nestings, 200 draws each, went through the same calibration pipeline, searching the ρ profile, cost inversion, and merger simulation. First (A), every ASIN has assigned to one of five nests uniformly at random, second (B) same group sizes as the CLIP cluster has been kept but the labels are mixed, which keeps the CLIP nest size distribution but severs the product to embedding link.

Table 9 and Figure 5 differentiate the generic from the real contribution.

The cost-side repair is generic. Randomly assigned ASIN nests push the negative marginal cost rate down to a median of 0.9%; they match or better CLIP's 1.7% in 96% of draws. This is what Appendix D proves – crossing ownership is what does the work. The observed 8.6% to 1.7% drop is real and it matters, but it cannot distinguish CLIP from noise on its own.

The demand fit is not generic. It is shown that M3's RSS drop of 8.1% beats 95% of the random partitions (median: 5.3%), and its $\rho^* = 0.67$ is higher than 95.5% of them. Because a nesting only reduces residual variance when the ASINs grouped together actually share demand shocks, embedding-identified products can do, while randomly assigned ones mostly add noise.

The merger effect is not generic and this is decisive. Every one of the 200 random draws puts hello's price effect above CLIP's +0.40%; the draws range from +2.5% at the minimum to +10.2% at the maximum (median +5.9%, a fourteen-fold gap; Table 9 reports the narrower 5–95% band). The mechanism is economic, not statistical. Colgate sells numerous ASINs. If these were assigned randomly across five nests, hello's nest will almost always capture a few. This creates a simulated hello to Colgate diversion without any underlying economic basis, and the apparent effect inflates accordingly.

On the other hand, CLIP-nested M3 manages to keep Colgate's entire product line within a single cluster, distinctly separate from hello's natural/specialty cluster. Consequently, any remaining diversion between the merging parties genuinely reflects actual product differentiation. Placebo B further emphasizes this point: permuting the CLIP labels keeps nest sizes constant, yet the negative-MC rate still decreases (median 1.6%) while the hello effect still inflates (median +5.2%). This demonstrates that CLIP's effectiveness lies in the accurate assignment of products to nests, rather than merely their sizes.

Nor is a hand-coded classification a substitute. As a stand-in for an analyst reading listings, we ran a keyword classifier on the raw title and benefit text, employing terms like "kids," "sensitivity," "whitening," "natural," and "standard." This classifier was run in two tie-breaking orders, as most listings contain multiple benefit claims, and it followed the same pipeline as other methods. However, these keyword-based classifications performed worse than a median random partition in terms of demand fit. The RSS drops were 4.2% and 4.4% compared to the random median of 5.3%. Both variants also estimated hello's price effect to be above +4.5%.

The "natural-first" variant provided the most instructive failure. While it successfully separated hello from the Colgate brand, it incorrectly grouped thirteen Colgate-owned Tom's of Maine listings with hello, due to their shared "natural" keywords. In contrast, CLIP assigned hello, Colgate, and Tom's of Maine to three distinct clusters. The embeddings recognized that hello's bright, design-led positioning differs significantly from Tom's of Maine's earth toned, health store positioning. This distinction, accurately identified by CLIP, resulted in the largest improvement in demand fit.

To sum up, these results precisely state the contribution: While cross-firm nesting can rectify the cost side, only CLIP achieves this while simultaneously outperforming 95% of random partitions in fitting demand and providing a merger counterfactual free of the spurious diversion introduced by those partitions. Brand nesting, at the other extreme, separates the merging parties but leaves the cost side unaddressed (Appendix D). A keyword classification performs no better, fitting worse than the median random draw and co-nesting "hello" with Colgate's natural subsidiary. Of the nesting rules considered, only CLIP satisfies all three criteria simultaneously.

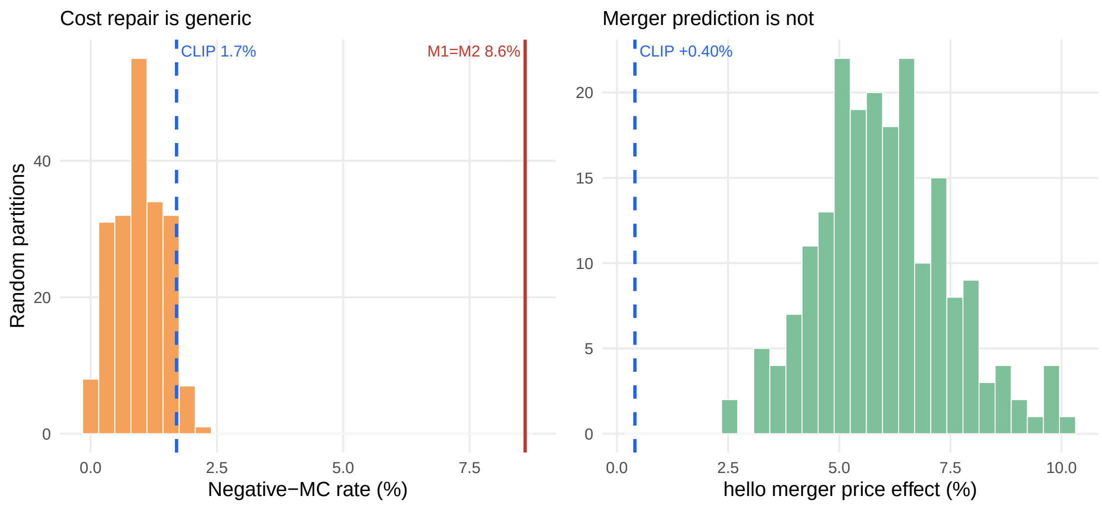

Figure 5: Placebo nesting tests (200 random cross-firm partitions). Left: random nests reach CLIP's negative-MC rate (blue dashed) or lower in 96% of draws. Right: CLIP's hello effect of +0.40% lies below all 200 draws.

Table 9: Placebo and Baseline Nesting Tests: CLIP vs Random, Shuffled, and Hand-Coded Partitions

Metric	CLIP (real)	Random A (median [5–95%])	Shuffled B (median)	Keywords (nat.-first)
ρ^*	0.67	0.53 [0.39, 0.66]	0.41	0.44
RSS drop (%)	8.10	5.30 [3.00, 7.91]	4.00	4.40
Neg-MC (%)	1.70	0.90 [0.20, 1.70]	1.60	1.90
hello effect (%)	**0.40**	**5.87 [3.80, 8.73]**	**5.15**	**5.52**
Colgate effect (%)	0.02	0.41 [0.26, 0.59]	0.39	0.08

Notes:
200 draws each: Random A assigns every ASIN to one of five nests at random; Shuffled B permutes the CLIP labels. All run through the identical pipeline as M3. Bold row: CLIP's +0.40% lies below all 200 draws (min +2.5%).

4.7 How much of the Result Driven by Brand?

The placebo tests show that explanatory power of CLIP nesting on substitution patterns is can not be achieved with an random cross firm partition. However, since the embeddings are built from the whole listing, and the brand name and logo are part of what a shopper sees, it is reasonable ask whether the CLIP clusters are simply brand groups under another name. Three diagnostics address this question, and the honest answer has two parts: while the clusters are strongly correlated with brands, but this correlation is not what the result solely depends on.

4.7.1 Where the nests sit relative to ownership

Table 10 breaks down the panel by CLIP cluster. The most important column is `n_firms`: the average CLIP cluster spans 2.80 distinct owning firms, compared to exactly 1.00 for brand nests by construction. This property is what Appendix D states is needed on the cost side, and the table demonstrates where the repair actually occurs. Two clusters—cluster 2, dominated by Colgate, and cluster 5, the heterogeneous "Other" group with 7 firms in it – and between them they account for almost all 80 observations where the implied marginal cost is negative under plain logit but non-negative under CLIP nesting. The single-firm clusters (3 and 4) show no repair, which aligns with the mechanism: without a rival firm within the nest, there is no new substitution for ρ to reallocate. The M1 and M2 columns are identical across all clusters. This is not a data coincidence; it reflects the Appendix D identity holding true for each cluster, as brand nests never cross an ownership boundary.

Table 10: Ownership Span and Negative-Cost Rate by CLIP Cluster

Cluster	Obs	ASINs	Firms	Dominant brand	Share	M1	M2	M3	Repaired
1	251	29	2	Sensodyne	39%	1.99%	1.99%	1.99%	0
2	254	33	3	Colgate	88%	14.17%	14.17%	3.94%	26
3	216	29	1	Crest	100%	2.31%	2.31%	2.31%	0
4	85	13	1	Tom's of Maine	100%	0.00%	0.00%	0.00%	0
5	358	55	7	Other	54%	15.08%	15.08%	0.00%	54

Notes:
Firms counts distinct owners under the pre-merger structure of Section 3.6. M1/M2/M3 give the negative-marginal-cost share; Repaired counts observations with mc < 0 under M1 and mc >= 0 under M3. Source: scripts/cluster_ownership_diagnostic.R.

4.7.2 Removing brand from the embedding space

If the success of the embedding clusters were purely independent of brand effects (which they clearly are not), one would expect new clusters built on the variance not explained by brands (residuals) to yield similarly discriminative results. Regressing each principal component on brand dummies shows that brand accounts for an average of 72.4% of the variance in the five components (the variance-weighted figure is 73.6%; component-specific values are presented in Appendix E). Clustering the residuals of those regressions yields an adjusted Rand Index against brand of -0.0008, compared with 0.62 for the CLIP clusters themselves, confirming that brand has indeed been removed.

Table 11 puts this post-adjustment clustering back into the same test pipeline. The negative-cost repair survives: the rate is 1.5% against CLIP's 1.7%. But demand fit falls, the RSS drop declining from 8.1% to 5.7%, and the simulated hello merger price effect jumps from +0.40% to +5.56%.

The reason behind this mechanism becomes clear in the diversion ratios column. Since these brand-neutralised clusters fail to sufficiently distinguish between hello and Colgate, they result in Colgate being included among hello's in-nest competitors. The diversion ratio rises from 0.41% to 5.95%. With an average of 6.00 firms per nest, this partition repairs the cost side, but it does so at the price of a worse demand fit and a substantially inflated merger price effect.

Table 11: Nesting on the Brand-Residual Embedding Space

Metric	Brand-orthogonal	CLIP
rho*	0.49	0.67
RSS drop (%)	5.70	8.10
neg-MC (%)	1.50	1.70
hello effect (%)	5.56	0.40
Colgate effect (%)	0.47	0.02
diversion hello->Colgate (%)	5.95	0.41
mean firms per nest	6.00	2.80
ARI vs brand	-0.00	0.62

Notes:
Joint PCs residualised on brand dummies at ASIN level, then k-means with K = 5 on the residuals (seed 42). Pipeline otherwise identical to M3. Diversion is share-weighted from hello ASINs over pre-merger quarters. Source: scripts/brand_orthogonal_nesting.R.

4.7.3 Absorbing brand with fixed effects

As a final and necessary test, we absorbed the fixed brand effects to measure whether the information captured by the CLIP components disappears. Table 12 presents these results.

Two important observations. First, the brand fixed effects increase the ρ^* value of the brand-nested model from 0.35 to 0.55, more than doubling the improvement in model fit (RSS drop) from 3.1% to 6.88%, and yet it has not altered the negative cost ratio at all: 8.59% in the non-nested model and 8.59% in the brand-nested model.

As long as the nests remain within ownership boundaries, improvements in demand fit cannot correct the cost side (Appendix D). The merger effect does shift slightly (+1.233% against +1.350%), because the proposition pins markups at the observed shares, not the share functions, so a different ρ traces a different counterfactual equilibrium (Section 3.6).

Second, adding brand fixed effects to the CLIP model had virtually no impact on the results: ρ^* moves from 0.67 to 0.64, the negative cost ratio from 1.7% to 1.80%, and the hello merger effect from +0.40% to +0.436%. Although the variance across brands was fully absorbed by the dummy variables, 4 of the five CLIP components remained significant at the 5% level (Appendix E). This clearly demonstrates that the components do not merely replicate brand identity.

Table 12: Specifications with Brand Fixed Effects

Specification	ρ^*	RSS drop	Neg. MC	hello Δp	Colgate Δp
brand FE, no nesting	0.00	0.00%	8.59%	+1.233%	+0.104%
brand FE + brand nesting	0.55	6.88%	8.59%	+1.350%	+0.105%
brand FE + CLIP nesting	0.64	8.70%	1.80%	+0.436%	+0.022%

Notes:
Eleven brand_grouped_50 dummies added to the design matrix; is_other dropped as collinear. Each row keeps its own rho profile, cost inversion and merger simulation. Source: scripts/brand_fe_variants.R.

5 Discussion and Limitations

All of the empirical studies and tests presented above and in the previous sections are intended to establish transparency, traceability, and a scientific design for each specific case. And all of them have shown that the method should not be used to make further claims without additional research. It is essential to mention certain limitations of the study:

To what extent the "negative marginal cost correction" reflects the soundness of the model: It is clear that the embedding structure may offer an advantage over the standard demand estimation approach in the literature. The "inflated markup" problem has long necessitated various assumptions in the literature.

> Negative marginal costs would cause mergers to have unrealistic welfare effects, so we assume throughout that all shares are small enough that all marginal costs are positive. With all product prices equal to unity, as we generally assume, avoiding negative marginal costs requires $\max\{s_j\} < (\beta - 1)/(\beta - \varepsilon)$, and there can be a negative marginal cost only if demand is inelastic.
>
> — Werden and Froeb (1994, p. 411, n. 7)

In their notation β is the cross-elasticity parameter governing substitution among the inside goods and ε the aggregate elasticity of demand for those goods; neither corresponds to the β that denotes the characteristic coefficients elsewhere in this thesis. At this point, it would not be wrong to argue that embedding nesting serves as a practical empirical solution rather than a mere assumption. However, the contribution remains generic, as explained in detail earlier in Section 4.6.

Cost recovery via demand: Since no cost-side variables providing sufficient variation for the online-retail panel data could be identified, costs were recovered via demand. This significantly limits the reliability of the estimated negative marginal costs.

No-efficiencies assumption: The "no efficiencies" assumption, established to isolate market power, completely disregards the marginal cost advantages that the merger in question could create.

Sparse purchase counts: As mentioned earlier, one of the major limitations of the study is the very limited direct access to online retail data. The resulting panel data showed 2,528 toothpaste purchases across 1,164 ASIN×quarter cells. The median cell records one purchasing household, and 53.9% of cells hold exactly one purchase. Although the applied filtering and methodological approaches (minimum three-quarter presence, quarterly aggregation, and a focus on cross-model comparisons) mitigate the impact of this limitation, the results should be interpreted with caution.

Static embeddings and cluster count: Embeddings are extracted once per product, which is appropriate for listings that change little, but would miss repositioning. The number of clusters is a free parameter, set to $K = 5$ on the negative-cost criterion; Appendix A reports $K = 2$ to 10 and shows the qualitative ranking is preserved but not robust in K – at $K = 4$ the negative-cost rate rises to 3.0% and hello's effect to +3.96%, which falls inside the placebo band. Cluster separation alone would favour $K = 9$, where the average silhouette peaks at 0.403 against 0.388 at $K = 5$. This is a limitation the diagnostics in Section 4.7 are designed to expose rather than hide.

External validity: This entire study was conducted on a single platform, a single category, and a single merger.

6 Conclusion

This paper documents the potential advantages of clustering logic based on unstructured data in explaining substitution effects. By replacing the nested logit models based on brand identity with product embeddings extracted from text and images, it demonstrates an increase in the demand fit (Table 4), better-recovered marginal costs on the supply side, and a significantly different merger price effect.

The contribution is entirely methodological and should be examined in two parts. In previous studies that inspired this work, representation learning was used as continuous product attributes in demand estimation (Compiani et al., 2026; Magnolfi et al., 2025). Here, however, they were primarily used to define the "partition." When embeddings are used to partition products into nests, nothing changes in the Berry (1994) estimator, and the entire pipeline remains a linear optimization problem that runs in seconds. In regulatory environments, the simplicity and interpretability of the structure may provide a compliance advantage over methodological advantages.

When it comes to the second part of the contribution: because nesting structures are normally chosen on institutional priors, using the repair of economically implausible results offers a different way to distinguish a good nesting from a bad one. It is a criterion that the demand side alone cannot deliver: an inversion that overstates markups reveals itself in costs below zero, independent of fit statistics.

Considering the results of placebo and brand contribution tests, embedding reproduces the product's position in a more discriminating manner, alongside the brand's significant contribution. Brand clustering can only form clusters based on brand effects and cannot cross ownership boundaries. Brand-residual nesting, on the other hand, completely cuts off ownership and loses the ability to position products. Only full embedding (brand-inclusive) can achieve both.

As a final word, today, the tools for credible merger simulation do not have to be limited to structured product data. Product images and listing texts, which are observable for any online marketplace at essentially zero cost to practitioners, along with a pre-trained encoder and a standard nested model, are sufficient to incorporate that information into the tools that are already in use by antitrust practitioners. The entire analysis and code are reproducible from a single document and a public repository, which is intended to make the method a useful toolkit in the future settings where it would be needed.

Data and Code Availability

All data, code, and the R Markdown source of this paper are openly available at `https://github.com/brkuzn/clip-amazon-toothpaste-market`. Knitting the source document re-runs the entire pipeline – the joint PCA, the ρ grid searches, and the Bertrand–Nash equilibria – and regenerates every table and figure from the included data. The repository also contains the CLIP extraction notebook, the joint-PCA construction script, and reference copies of all outputs; reproduction of the shipped joint principal components is byte-exact. Raw household-level purchase records with demographics are not redistributed and are available from the Open E-commerce 1.0 repository (Berke et al., 2024).

References

Bajari, P., Nekipelov, D., Ryan, S. P., & Yang, M. (2015). Machine learning methods for demand estimation. *American Economic Review, 105*(5), 481–485.

Berke, A., Calacci, D., Mahari, R., Yabe, T., Larson, K., & Pentland, S. (2024). Open e-commerce 1.0, five years of crowdsourced U.S. Amazon purchase histories with user demographics. *Scientific Data, 11*, 491. https://doi.org/10.1038/s41597-024-03329-6

Berry, S. T. (1994). Estimating discrete-choice models of product differentiation. *RAND Journal of Economics, 25*(2), 242–262.

Berry, S. T., Levinsohn, J., & Pakes, A. (1995). Automobile prices in market equilibrium. *Econometrica, 63*(4), 841–890.

Bijmolt, T. H. A., van Heerde, H. J., & Pieters, R. G. M. (2005). New empirical generalizations on the determinants of price elasticity. *Journal of Marketing Research, 42*(2), 141–156.

Björnerstedt, J., & Verboven, F. (2016). Does merger simulation work? Evidence from the Swedish analgesics market. *American Economic Journal: Applied Economics, 8*(3), 125–164.

Cardell, N. S. (1997). Variance components structures for the extreme-value and logistic distributions with application to models of heterogeneity. *Econometric Theory, 13*(2), 185–213.

Colgate-Palmolive Company (2020). *We Are Colgate: 2020 Annual Report*. Colgate-Palmolive Company.

Compiani, G., Morozov, I., & Seiler, S. (2026). Demand estimation with text and image data. *The RAND Journal of Economics*. https://doi.org/10.1111/1756-2171.70052

Conlon, C., & Gortmaker, J. (2020). Best practices for differentiated products demand estimation with PyBLP. *RAND Journal of Economics, 51*(4), 1108–1161.

Conlon, C., & Mortimer, J. H. (2021). Empirical properties of diversion ratios. *RAND Journal of Economics, 52*(4), 693–726.

DeSouza, S. A. (2009). *The Antitrust Mixed Logit Model*. Working Paper No. 04, Graduate School of Economics (CAEN), Universidade Federal do Ceará, Fortaleza, Brazil.

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., . . . & Houlsby, N. (2021). An image is worth 16x16 words: Transformers for image recognition at scale. *International Conference on Learning Representations (ICLR).*

Draganska, M., & Jain, D. C. (2006). Consumer preferences and product-line pricing strategies: An empirical analysis. *Marketing Science, 25*(2), 164–174.

Farrell, J., & Shapiro, C. (1990). Horizontal mergers: An equilibrium analysis. *American Economic Review, 80*(1), 107–126.

Farrell, J., & Shapiro, C. (2010). Antitrust evaluation of horizontal mergers: An economic alternative to market definition. *The B.E. Journal of Theoretical Economics, 10*(1), Article 9.

Gandhi, A., & Houde, J.-F. (2019). Measuring substitution patterns in differentiated-products industries. NBER Working Paper 26375.

Grigolon, L., & Verboven, F. (2014). Nested logit or random coefficients logit? A comparison of alternative discrete choice models of product differentiation. *Review of Economics and Statistics, 96*(5), 916–935.

Hausman, J. A. (1996). Valuation of new goods under perfect and imperfect competition. In T. F. Bresnahan & R. J. Gordon (Eds.), *The Economics of New Goods* (pp. 209–237). University of Chicago Press.

Hotelling, H. (1933). Analysis of a complex of statistical variables into principal components. *Journal of Educational Psychology, 24*(6), 417–441.

Huang, D., Rojas, C., & Bass, F. (2008). What happens when demand is estimated with a misspecified model? *The Journal of Industrial Economics, 56*(4), 809–839.

Lloyd, S. P. (1982). Least squares quantization in PCM. *IEEE Transactions on Information Theory, 28*(2), 129–137.

Magnolfi, L., McClure, J., & Sorensen, A. T. (2025). Triplet embeddings for demand estimation. *American Economic Journal: Microeconomics, 17*(1), 282–307.

McFadden, D. (1974). Conditional logit analysis of qualitative choice behavior. In P. Zarembka (Ed.), *Frontiers in Econometrics* (pp. 105–142). Academic Press.

McFadden, D. (1978). Modelling the choice of residential location. In A. Karlqvist, L. Lundqvist, F. Snickars, & J. W. Weibull (Eds.), *Spatial Interaction Theory and Planning Models* (pp. 75–96). North-Holland.

Morrow, W. R., & Skerlos, S. J. (2011). Fixed-point approaches to computing Bertrand-Nash equilibrium prices under mixed-logit demand. *Operations Research, 59*(2), 328–345.

Nevo, A. (2000a). A practitioner's guide to estimation of random-coefficients logit models of demand. *Journal of Economics & Management Strategy, 9*(4), 513–548.

Nevo, A. (2000b). Mergers with differentiated products: The case of the ready-to-eat cereal industry. *RAND Journal of Economics, 31*(3), 395–421.

Nevo, A. (2001). Measuring market power in the ready-to-eat cereal industry. *Econometrica, 69*(2), 307–342.

Peters, C. (2006). Evaluating the performance of merger simulation: Evidence from the U.S. airline industry. *Journal of Law and Economics, 49*(2), 627–649.

Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., ... & Sutskever, I. (2021). Learning transferable visual models from natural language supervision. *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 8748–8763.

Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics, 20*, 53–65.

Ruiz, F. J. R., Athey, S., & Blei, D. M. (2020). SHOPPER: A probabilistic model of consumer choice with substitutes and complements. *Annals of Applied Statistics, 14*(1), 1–27.

Small, K. A., & Rosen, H. S. (1981). Applied welfare economics with discrete choice models. *Econometrica, 49*(1), 105–130.

Train, K. E. (2009). *Discrete Choice Methods with Simulation* (2nd ed.). Cambridge University Press.

U.S. Department of Justice & Federal Trade Commission (2010). *Horizontal Merger Guidelines*. Washington, DC. Issued August 19, 2010.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems, 30*, 5998–6008.

Werden, G. J., & Froeb, L. M. (1994). The effects of mergers in differentiated products industries: Logit demand and merger policy. *Journal of Law, Economics, & Organization, 10*(2), 407–426.

Appendix

Appendix A: K Sensitivity Analysis

Table 13: K Sensitivity Analysis: CLIP Cluster Count K = 2–10

K	WCSS	Silhouette	ρ^*	RSS Drop	Neg. MC%	hello Δp%	Colgate Δp%
2	1042.0	0.199	0.60	6.1%	0.5%	1.71%	0.015%
3	836.7	0.256	0.66	7.8%	0.9%	3.49%	0.013%
4	641.2	0.330	0.57	5.7%	3.0%	3.96%	0.041%
5	**456.5**	**0.388**	**0.67**	**8.1%**	**1.7%**	**0.40%**	**0.018%**
6	365.6	0.387	0.60	6.7%	3.4%	0.59%	0.044%
7	322.0	0.368	0.48	4.6%	4.4%	0.69%	0.055%
8	279.3	0.400	0.37	2.9%	4.7%	0.82%	0.067%
9	247.7	0.403	0.35	2.4%	4.7%	1.26%	0.177%
10	230.7	0.373	0.40	3.8%	4.6%	0.79%	0.065%

Notes: K-means on 5 CLIP PCs (seed 42, 50 restarts). WCSS = within-cluster sum of squares. Silhouette = average silhouette width (higher = better separation). ρ^* = argmin RSS on grid [0, 0.01, …, 0.90]. RSS drop = $100 \times (RSS_0 - RSS^*)/RSS_0$. Neg. MC = share of ASINs with negative implied marginal cost. hello / Colgate Δp = average post-merger price effect in percent (2020Q1–2022Q3). **Bold/green row = selected specification (K=5).**

Appendix B: Model Fit — RMSE and MAE

Table 14: Model Fit: RMSE / MAE vs. Observed Prices (Post-Merger Quarters 2020Q1–2022Q3)

	RMSE / MAE (USD)		
Brand / Scope	M1: Logit	M2: Brand-Nested	M3: CLIP-Nested
All	**1.558 / 0.743**	**1.556 / 0.746**	**1.564 / 0.810**
hello	1.579 / 0.836	1.580 / 0.837	1.533 / 0.799
Colgate	1.144 / 0.559	1.142 / 0.560	1.149 / 0.652

Notes: Each cell shows RMSE / MAE in USD against observed transaction prices. Nash (merger) prices are used as the model prediction. “All” = all 847 ASIN $\times$ quarter observations in the post-merger period (2020Q1–2022Q3); hello = 35 obs.; Colgate = 172 obs. Pre-merger marginal costs are held fixed in both Nash simulations.

Appendix C: Robustness Checks

Table 15: Robustness: Market Size, Calibration Level, and Cost Window

Scenario	λ	α	ρ^*	RSS drop	Neg. MC %	hello Δp	Colgate Δp
M1 baseline	**13.93**	**-0.310**	**0.00**	**0.0%**	**8.6%**	**+1.233%**	**+0.104%**
M1, market size x2	27.86	-0.310	0.00	0.0%	8.4%	+0.634%	+0.054%
M1, market size x5	69.65	-0.309	0.00	0.0%	8.4%	+0.258%	+0.022%
M1, market size x10	139.30	-0.309	0.00	0.0%	8.4%	+0.130%	+0.011%
M3 baseline	**13.93**	**-0.310**	**0.67**	**8.1%**	**1.7%**	**+0.404%**	**+0.018%**
M3, market size x2	27.86	-0.310	0.67	8.1%	1.6%	+0.211%	+0.010%
M3, market size x5	69.65	-0.309	0.67	8.1%	1.6%	+0.087%	+0.004%
M3, market size x10	139.30	-0.309	0.67	8.1%	1.6%	+0.044%	+0.002%
M2, alpha recalibrated at rho*	13.93	-0.212	0.35	3.1%	20.7%	+1.810%	+0.157%
M3, alpha recalibrated at rho*	13.93	-0.111	0.67	8.1%	36.3%	+0.944%	+0.057%
M3, MC from 2019Q4 only	13.93	-0.310	0.67	8.1%	1.7%	+0.440%	+0.022%

Notes: Bold rows are the baseline specifications. **Market size** rows multiply λ (and hence market size M_t) by 2, 5, and 10; α is recalibrated to the -2.62 target at $\rho = 0$ within each scenario. Fit and the negative-cost rate are essentially unchanged; absolute merger effects shrink mechanically as more diversion flows to the outside good, while the M1-to-M3 ratio of hello effects is stable. **Alpha recalibrated at ρ^*** rows hold the baseline profiled ρ^* fixed and rescale α so the model's own implied median elasticity equals -2.62; the resulting small $|\alpha|$ inflates Bertrand markups and drives negative-cost rates far above the plain-logit level, indicating that ASIN-level demand must be more elastic than the aggregate benchmark to rationalize observed prices. **MC from 2019Q4 only** fixes marginal costs at the last pre-merger quarter instead of the pre-merger average.

Appendix D: Equivalence of plain-logit and brand-nested markups

Proposition. Consider nested-logit demand with nesting parameter $\rho \in [0,1)$ and Bertrand competition, and suppose the nest partition refines the ownership partition: every nest g is wholly owned by a single firm. Then the multiproduct Bertrand markups solving the first-order conditions at given shares are identical to the plain-logit ($\rho = 0$) markups: for every product j of firm f, $p_j - mc_j = 1/(|\alpha|(1-S_f))$, where $S_f = \sum_{k\in f} s_k$ is the firm's market share.

Proof. Write $x_k = p_k - mc_k$. Firm f's first-order condition for product j is $s_j + \sum_{k\in f} x_k\, \partial s_k/\partial p_j = 0$. Under plain logit, substituting the share derivatives and dividing by αs_j gives $x_j - \sum_{k\in f} s_k x_k = -1/\alpha$, whose solution is uniform within the firm: $x_j = \mu_f$ with $\mu_f(1-S_f) = -1/\alpha$. Under nested logit with j in nest $g \subseteq f$, the same substitution gives

$$\frac{x_j}{1-\rho} - \frac{\rho}{1-\rho}\sum_{k\in g} s_{k|g}\, x_k - \sum_{k\in f} s_k x_k = -\frac{1}{\alpha}.$$

Try the uniform candidate $x_k = \mu_f$ for all $k \in f$. Because the entire nest g belongs to firm f, the within-nest shares sum to one, $\sum_{k\in g} s_{k|g} = 1$, so the first two terms collapse to $\mu_f\left(\frac{1}{1-\rho} - \frac{\rho}{1-\rho}\right) = \mu_f$, and the condition reduces to $\mu_f(1-S_f) = -1/\alpha$ — the plain-logit equation. Since the system $(\Omega \circ \Delta)\,\mathbf{x} = -\mathbf{s}$ is invertible in our data, this uniform solution is the unique solution, and it does not depend on ρ. □

The empirical counterpart is exact: across all 1,164 observations, M1 and M2 marginal costs agree to machine precision. The equivalence breaks whenever a nest spans products of different owners — the CLIP clusters do, which is why M3 recovers different (and far more plausible) costs.

Appendix E: Brand-Absorption Diagnostics

Supporting detail for Section 4.7. Both tables are generated from committed result files; the scripts that produce them are named in the notes and can be run standalone from the repository root.

Table 16: Share of Each Joint Principal Component Explained by Brand Dummies

Component	Brand R^2
joint_pc1	0.8701
joint_pc2	0.7642
joint_pc3	0.8197
joint_pc4	0.7247
joint_pc5	0.4429

Notes:
ASIN-level OLS of each component on a full set of brand_grouped_50 dummies. The variance-weighted average across the five components is 0.736, so the brand-residual space used in Section 4.7 retains 26.4% of the original variance. Source: scripts/brand_orthogonal_nesting.R.

Table 17 reports the joint-PC coefficients once brand is absorbed. The joint explanatory power survives – 4 of five components remain significant at the 5% level under CLIP nesting – but the individual loadings are not invariant. `joint_pc3` loses significance (t = 1.52), and two loadings change sign relative to the baseline specification while `joint_pc2` grows several-fold. The defensible claim is therefore about the components jointly, not about the sign or magnitude of any single loading, which should not be read as a structural parameter in either specification.

Table 17: Joint-PC t Statistics and Loadings under Brand Fixed Effects

Component	No nesting	Brand nesting	CLIP nesting	Baseline	Under brand FE
joint_pc1	-3.939	-4.247	-4.273	-0.3280	-0.3728
joint_pc2	9.522	9.883	9.918	+0.2103	+0.8318
joint_pc3	1.716	1.430	1.519	-0.2854	+0.1124
joint_pc4	7.755	7.848	8.058	+0.2707	+0.5485
joint_pc5	2.434	2.806	2.677	-0.1745	+0.1593

Notes:
t statistics on the five joint principal components in each brand-fixed-effects specification. The last two columns compare each component's loading in the baseline M3 specification with its loading once brand is absorbed. Critical value at the 5% level is 1.96. Source: scripts/brand_fe_variants.R.